

Munkres 3

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1 Introduction

Is the boundary of a convex subset of $\mathbb{R}^2$ connected?

23.1. Let X, Y be connected spaces. Show that if A and B form a separation of $X - Y$, then $A \cup Y$ and $B \cup Y$ are connected.

Let X, Y be spaces with the order topology. Suppose $h : X \rightarrow Y$ is a homeomorphism. Is h an order-preserving or order-reversing map?

If $\mathcal{A}$ is a collection of disjoint open intervals in $\mathbb{R}$, then $\mathcal{A}$ is countable. For each $I_\alpha \in \mathcal{A}$, pick rational $q_\alpha \in I_\alpha$, so $\mathcal{A} \hookrightarrow \mathbb{Q}$.

25: 5, 7, 8, 10.

Is an arbitrary intersection of compact sets compact?

Let $\mathcal{A}$ be an open locally connected cover of X . Then X is locally connected.

Let $f : X \rightarrow Y$ be a closed continuous surjective map. Let Y be compact and suppose for all $y \in Y$, $f^{-1}(\{y\})$ is compact. Then X is compact.

In $\mathbb{R}^2$ let A be a bounded subset with diameter d . Does there exist a point $x \in \mathbb{R}^2$ such that $A \subset \overline{U}(x, d/2)$? Does this hold for general $\mathbb{R}^n$? No.

To show x is not a limit point of A , find U such that $U(x) \cap A = \{x\}, \emptyset$.